

Chapter 12

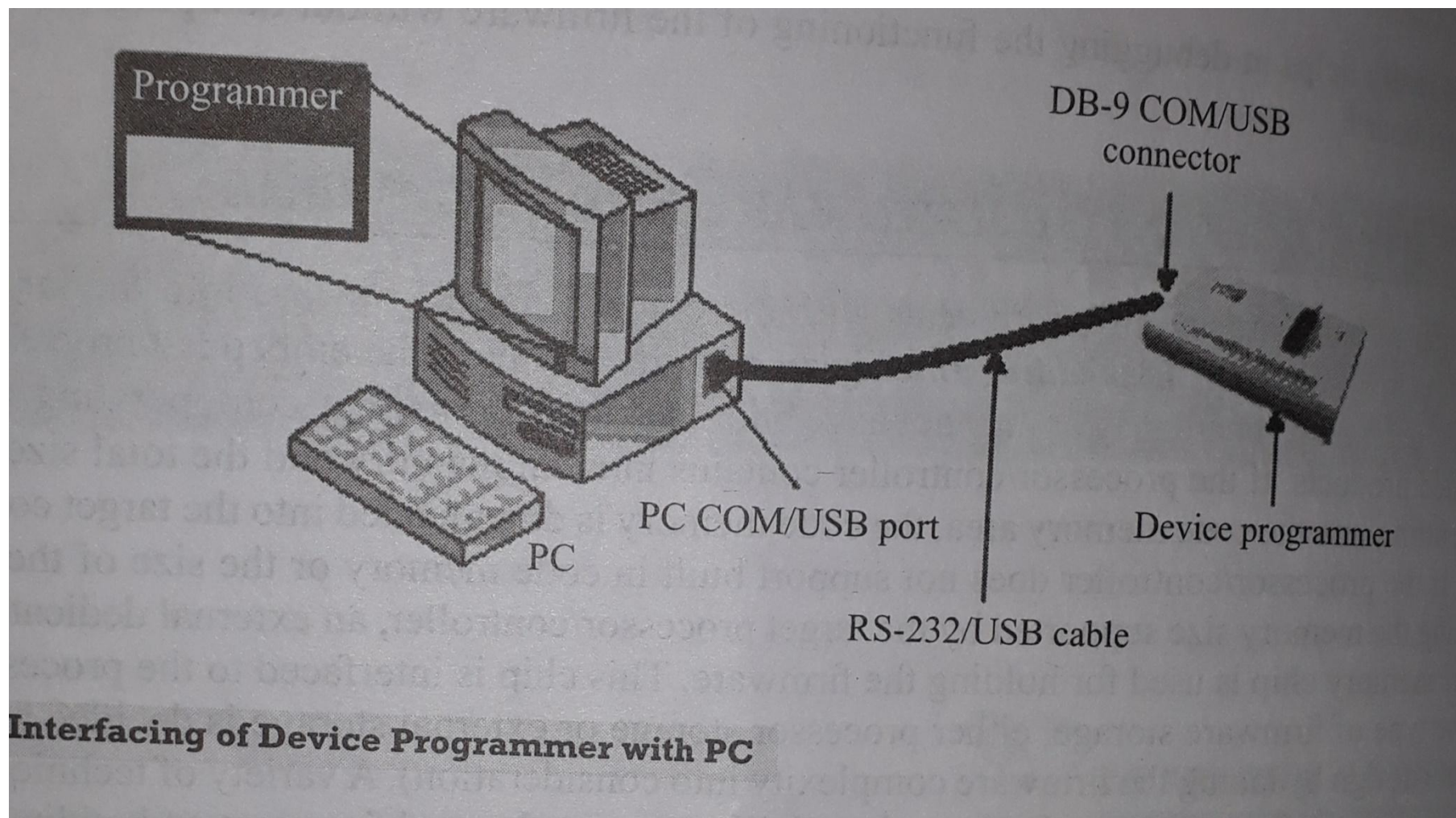
Integration and testing of Embedded Firmware

- **Integration of Hardware and Firmware**
 - It is the process of Embedding intelligence
 - For non-OS based processor, if sufficient code memory is available, then code is downloaded into target controller/processor.
 - If memory is not sufficient external memory (EPROM or Flash) is used for holding firmware

I. Firmware Embedding techniques for a non-OS based ES

1) Out of Circuit Programming

- ✓ It is performed outside the target board.
- ✓ The processor or memory chip into which the firmware needs to be embedded is taken out of target board and it is programmed with the help of programming device.
- ✓ The device contains ZIF (Zero Insertion Force) socket with locking pin to hold the device to be programmed.
- ✓ Programmer is interfaced to PC through RS232C/USB/parallel port interface.



The sequence of operations for embedding the firmware with a programmer are:

- Connect the programming device to specified port
- Power up the device
- Execute the programming utility on PC and ensure proper connectivity is established between PC and programmer.
- Unlock ZIF socket by turning the lock pin
- Insert the device to be programmed into open socket
- Lock the ZIF socket
- Select the device name from the list
- Load the hex file
- Program the device by program option
- Wait till the completion of programming operation
- Ensure programming is successful by checking status LED.
- Unlock ZIF socket and take the device out of programmer.

Disadvantages of out-of-circuit programming is:

- High development time
- whenever firmware is changed, the chip should be taken out of development board for re-programming
- This is tedious and prone to chip damages due to frequent insertion and removal
- Not suitable for batch production.

Hence, used for development of low volume products and proof of concept Development.

2) IN system Programming (ISP)

- Programming is done within the system (firmware is embedded into the target device without removing the target board.)
- Most flexible and easy way of firmware embedding.
- Pre-requisite is that the target device must have an ISP support.
- The communication between the target device and ISP utility is in serial format.
- The serial protocols are: JTAG or SPI or any other interface protocol.

ISP with SPI protocol

- On-chip EEPROM or FLASH memory is programmed through this interface.
- The primary I/O lines involved in SPI are: -
 - MOSI-Master out slave in
 - MISO-Master in slave out
 - SCK-System clock
 - SS- slave select
 - RST- reset of target device
 - GND- gnd of target device
- PC acts as master and target device acts as slave in ISP.
- The programmer data is sent to MOSI pin of target device and device acknowledgement is originated from MISO pin of device 8

1. Apply supply voltage between VCC and GND
2. Set RST pin to High state.
3. If a crystal is not connected across pins XTAL1 and XTAL2, apply 3MHz to 24MHz clock
4. Enable serial programming by sending programming enable serial instruction to MOSI pin
5. The code is programmed one byte at a time with appropriate write command.
6. Memory location can be verified by using read instruction for the data from MISO.
7. After successful programming the device, set RST to low and turn off chip power supply.

3) In application programming (IAP)

- Used for modifying selected portion of code memory.
- It is not a technique for first time embedding of user written firmware.
- Used for updating calibration data, look up table etc.
- The Boot ROM resident IAP instructions, performs various functions such as programming, erasing, and reading.
- A common entry point to these IAP routines is provided for interfacing them to end users application.
- Like a subroutine a function is called and returned to end user code.
- When a status bit is set, access to internal code memory is from Boot ROM.

4) Use of factory programmed chip

- Embedding of firmware into target can also be done during chip fabrication.
- Such chips are known as factory programmed chips.
- After the design of firmware, and operational stability is achieved, firmware files can be sent to chip fabricator to embed into code memory.
- Factory programmed chips are convenient for mass production. And it reduces product development time.
- It is not recommended to use factory programmed chips for development purpose where the firmware goes frequent changes.
- Factory programmed chips are bit expensive.

II. Firmware loading for operating system based devices

- OS based ES are programmed using ISP technique.
- OS based ES contains a special code called Boot loader Program which takes control of the OS and application firmware embedding and copying the OS image to the RAM of the system for execution.
- ‘Boot loader’ for such ES comes as pre-loaded or it can be loaded to the memory using the various interfaces supported like JTAG.
- The bootloader contains many driver initialization like UART, TCP/IP etc.
- Bootloader implements menu options for selecting source for OS image to load.
- Once communication is established then OS image downloaded to FLASH memory of target device.